# IS2203 Object-Oriented Programming

## Practical - 02

---

### ගමන්සර (GamanSara): Transforming Sri Lanka's Public Transport System

After successfully completing the initial development phase of the GamanSara transport management system, the development team has now started organizing transport-related information using structures, classes, and objects in C++.

During the previous phase, the system handled only simple inputs and outputs related to buses, passengers, and transport services. However, as the transport network expanded, developers realized that handling large amounts of transport information using separate variables was becoming difficult to manage.

To improve the system, the development team decided to group related transport information together using structures. Passenger details, bus information, and train details could now be stored in a more organized manner.

For example:

- A Passenger should contain passenger name, passenger ID, age, and destination.
- A Bus should contain bus number, route name, and seating capacity.

As development continued, the team introduced classes and objects to better represent real-world transport entities in the system. The developers also started using access specifiers and member functions to improve security and modularity.

The GamanSara development team is currently implementing these new OOP concepts to continue building a scalable smart transport management system for Sri Lanka.

1. Write a C++ program to declare and display the following bus details:

    Bus Number - 120
    Route Number - 138
    Average Fare -85.50
    Service Type -'L'

2. Write a C++ program to store the following passenger counts using an array and display all values using a loop.

    45, 38, 52, 41, 60

3. Write a C++ program using a structure called Passenger to store the following details:

    Passenger Name
    Passenger ID
    Age
    Destination

4. Write a C++ program using a structure called Bus with the following members:

busNumber
routeName
capacity
Create and display details of two buses.

5. Write a C++ program to create a class called Train.

Data Members: trainName, trainNumber, passengerCapacity
Member Functions: setTrainDetails(), displayTrainDetails()
Create one object and display train details.

6. Create a class called Ticket.

ticketNumber -public
fare -private
Try to access both variables inside the main function.
What happens? Explain the reason.

7. Identify errors, correct, and rewrite the program:

```
#include<iostream>
using namespace std;

class Passenger
{
public:

string name;
int age;

void display()
{
cout<<"Passenger Name:"<<name<<endl;
cout<<"Passenger Age:"<<age<<endl;
}
};

int main()
{
Passenger P1;

P1.name = "Kamal";
P1.age = 25;

P1.display();

return 0;
}
```

8. What is the difference between a structure and a class in C++?

9. What is the purpose of access specifiers in OOP?

10. Why are classes useful in developing large systems such as GamanSara?